INTERVIEW CASE STUDY · INTERNATIONAL INCENTIVES

Smarter incentives, built to land in a real organisation.

Turning a slow, meeting-heavy experiment process into a lean AI operating system that spends less, learns faster, and ships value in weeks, not quarters.

AI Systems, Personalization & Growth Manager · prepared for the Tripadvisor interview team

THE REFRAME

In a big company, the model is the easy 10%.

A recommendation engine is a weekend prototype. Getting it used, trusted, and safe across teams is the year of work. So I would spend my effort on the four things that actually block value.

MODEL · 10% THE 90% THAT DECIDES SUCCESS

Data & ETL

Investigate, clean and pipe the data so results can be trusted.

Integration

Plug into the warehouse, tracking and Braze, not a side tool.

UX & adoption

Embed AI where teams already work, so it gets used properly.

Change management

Small wins and coalition-building to earn resource and buy-in.

HOW TO READ THIS

Every question in the brief, answered and labelled.

Each slide carries a badge with the question it answers. Four green “in practice” slides sit next to the part they support, showing how it actually lands in a real org.

PART 1
Data reasoning

- **1.1** Segments + deeper analysis
- **1.1** Metrics I trust vs question
- **1.2** Extra data I'd request
- **1.3** The next three experiments

PART 2
AI systems design

- **2.1** End-to-end workflow
- **2.2** Which two to build first
- **2.3** Knowledge management
- **2.4** AI personas
- **2.5** Measuring success

PART 3
AI debugging

- **P3** 5+ reasons the rec failed
  + how I'd investigate each

PART 4
Stakeholders

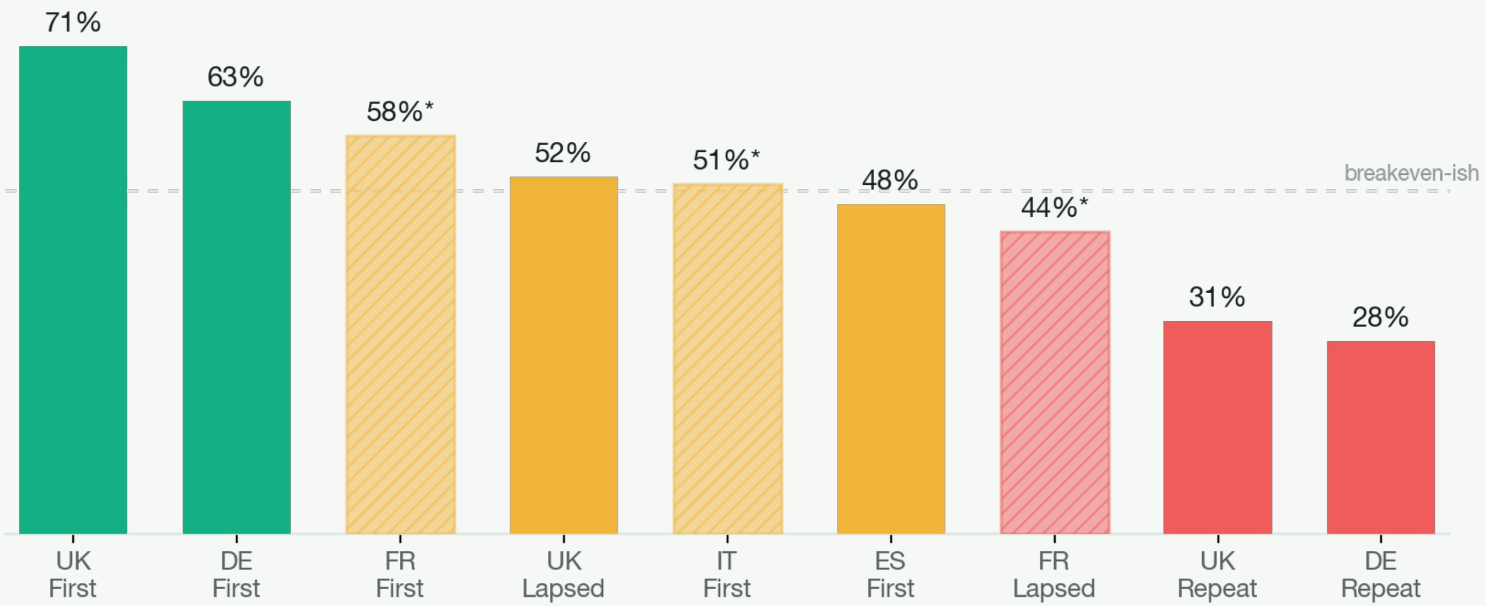
- **P4** Aligning Finance & the Lead
  ~300-word response
- → In practice: change mgmt

1.1

ANSWERING **Which segments create or destroy value?**

One flat rule is hiding two different businesses.

Read by incrementality, the pattern is clear: the newer the customer, the more the discount actually causes the booking. We pay the same 15% to people who need convincing and to people who were already coming.



VALUE CREATED

First purchase: UK 71%, DE 63% incremental. New customers we genuinely win.

VALUE DESTROYED

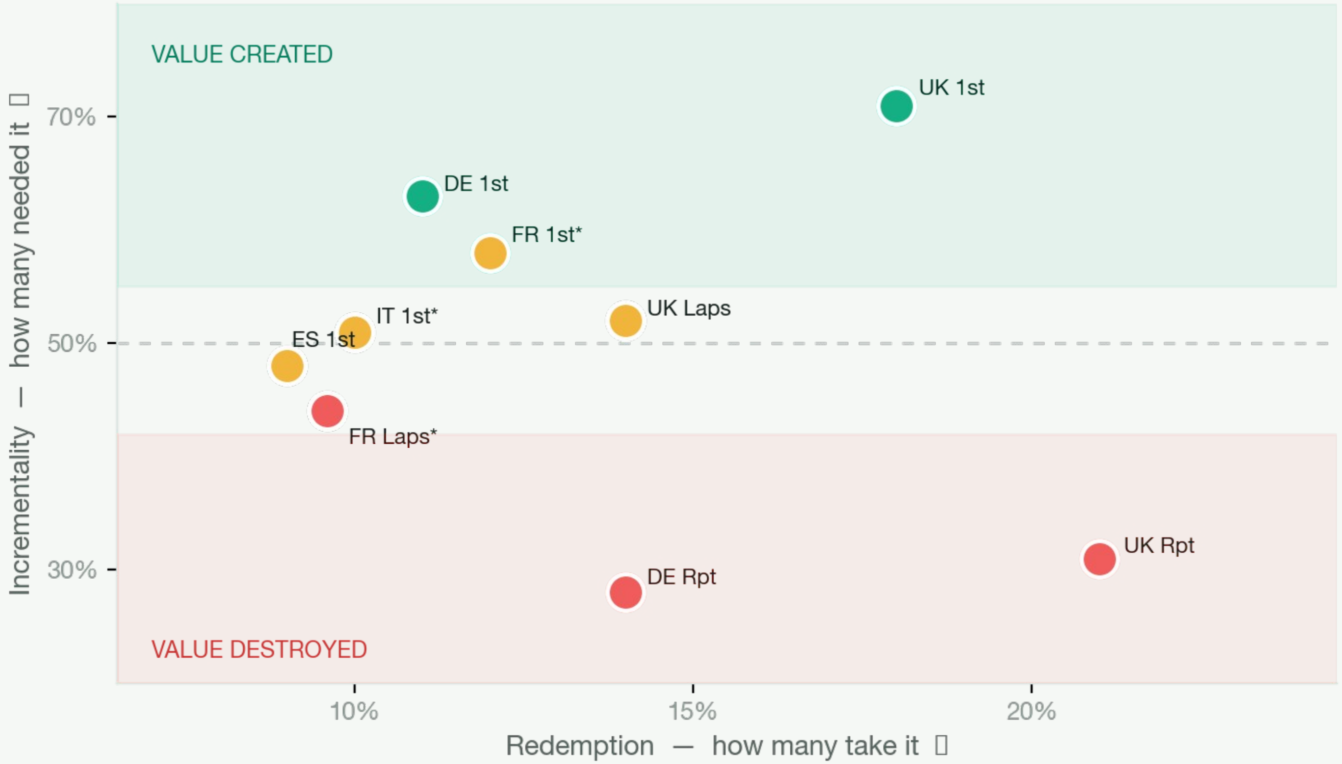
Repeat buyers: DE 28%, UK 31%. We subsidise people who would book anyway.

The metric that matters is incrementality, not redemption. UK repeat redeems the most and is nearly the least incremental.

1.1 ANSWERING **Going deeper: what advanced analysis reveals.**

Redemption is a vanity metric. Map it against value.

Plot every segment two ways: how many take the discount (redemption) vs how many actually needed it (incrementality). Winners sit high; the money leaks bottom-right.



THE ANGLE

High redemption looks like success. But UK Repeat redeems the MOST (21%) and is nearly the least incremental (31%). That's spend, not growth, hiding as a busy promo.

THE MATHS BEHIND IT

Cost / incremental booking = promo ÷ incrementality → repeat costs ~3× more.
Deadweight = promo × (1−incr) → ~70% of repeat spend wasted.

Next, with more data: iROAS & LTV:CAC per segment, CUPED, uplift modelling.

1.1 ANSWERING **Which metrics do you trust, and which would you question?**

I trust the metric. I question the measurement.

The apparent contradiction resolved: incrementality is the right metric (it's causal). But a metric is only as good as the experiment behind it, and two of these were built on broken experiments.

TRUST – THE METRIC TYPE

Incrementality is causal. A holdout gives the counterfactual (what people do with no offer), so it isolates cause, not correlation.
It beats redemption & CVR lift. Those are observational: they count what happened, not what we caused. Incrementality nets out the free-riders.
Where the holdout is clean (UK, DE), I'd act on it, it's the one number that separates real growth from subsidised habit.

QUESTION – THE MEASUREMENT

FR & IT incrementality (the same metric). Targeting issues broke randomisation, so the estimate is biased, not just noisy. Directional only.
No sample size or confidence interval. Can't separate signal from noise; a 71% on a tiny cell isn't the same as 71% on 10k.
Attribution traps. Redemption is self-selected; last-click over-credits the offer; pooling markets risks Simpson's paradox.

Six things I'd ask for, and how each changes the call.

I wouldn't finalise on this table alone. Each of these moves a specific decision, not just adds detail. If a data point can't change what I do, I don't wait for it.

Customer lifetime value

First-booking value is a snapshot; LTV is the prize.

Changes: A high-LTV first customer justifies a deeper discount, it's LTV:CAC, not first-order profit. Could flip 'marginal' ES/IT/FR into worth winning.

Contribution margin

Promo cost is clear; the profit it eats isn't.

Changes: Sets the real breakeven discount depth, and whether a “negative” verdict is mild or severe.

Sample size & confidence

No n or intervals are shown.

Changes: Decides whether I roll out, retest, or ignore a row. Urgent given the France/Italy flags.

Traffic source

A coupon-site redemption is low-incrementality by design.

Changes: Changes targeting: I may exclude deal-seeker channels rather than re-discount the same people.

Device & app usage

App / logged-in users skew loyal and repeat.

Changes: Changes eligibility: suppress the offer for high-frequency app users who convert without it.

Holdout integrity

The France/Italy targeting issue is the biggest unknown.

Changes: Decides whether those markets earn an experiment slot, or a measurement fix first.

1.3 ANSWERING **Recommend the next three experiments.**

Only three slots a quarter. Spend them on learning that pays.

01
Stop paying loyal customers

MARKET UK & Germany **AUDIENCE** Repeat buyers

Offer. Holdout: 15% vs 0% vs a non-cash perk.
Why. Repeat is the clearest value-destroyer (28–31%). Biggest, fastest saving for Finance.
Learn. How much repeat revenue is truly incremental.

Business risk. Some repeat bookings may be incremental. The holdout caps exposure; watch retention, not just next booking.

02
Fixed value vs percentage

MARKET Germany **AUDIENCE** First purchase

Offer. A/B €15 voucher vs 15%. Cost matched on a €102 basket.
Why. Test the qualitative hypothesis on the proven segment, don't assume it.
Learn. Whether framing lifts incrementality at equal cost.

Business risk. Framing effect may be small, but cost is matched, so the downside is a slot, not budget.

03
How shallow can we go?

MARKET Spain **AUDIENCE** First purchase

Offer. Depth test: 10% vs 15%, plus a €10-off-€75 threshold.
Why. Marginal (48%) but not flagged, lowest basket. Cleanest place to find the floor.
Learn. The discount elasticity that answers Finance directly.

Business risk. Under-discounting could slow acquisition. Measure downstream CLV, not just first bookings.

✋ **Deliberately not France or Italy.** Their data is flagged. Spending a scarce slot on numbers we can't trust would be the mistake. Fix the holdout first.

“Incrementality should be treated as directional.”

That one line in the brief is the whole job. France and Italy had targeting issues in the holdout, so their numbers cannot be trusted. You cannot automate on data you cannot trust, and this is not a rare event, it is the default state of data in a large org.

Raw tracking

Backend events, web & app, Braze sends

ETL & clean-up

Dedupe, fix attribution, single source of truth

Validated experiment data

Holdout integrity, sample sizes, confidence

AI can act

Only now is automation safe

Where I would start: a data investigation and clean-up sprint plus a re-run of the France/Italy holdout, before a single model ships. Realistically 60–70% of the effort. It is unglamorous, and it is the difference between an AI that helps and one that confidently misleads.

2.1 ANSWERING **Design the end-to-end AI workflow.**

One operating system, built on CRISP-DM.

The framework I used to ship a multi-agent system at Indeed Flex. Business and data understanding come **before** modelling, which is exactly what the ~95% of failed enterprise AI projects skip. The loop is what turns one experiment into a system.

1
Business understanding
The "is it worth it?" gate. Define the decision & metric. Sometimes the answer is a simple rule, not a model.

2
Data understanding
Trust the data first. The France/Italy lesson: no automation on data you can't trust.

3
Data preparation
ETL, clean, validated holdouts. The unglamorous 60–70%.

4
Modelling
Draft the experiment / offer. Simplest model that works, not the fanciest.

✓
Evaluation
Holdout, incrementality, significance + human sign-off before launch.

5
Deployment
Launch, monitor, write the learning brief.

↻ **It's a loop, not a line.** Deployment feeds the next Business-understanding step, so every experiment makes the system sharper. AI drafts and recalls; humans own the judgement; async gates replace the review meetings (most of the 60%).

Covers all six: Inputs (hypothesis) · Data sources (warehouse + evidence base) · AI capabilities (recall, draft, QA, summarise) · Human review (Evaluation gate) · Outputs (spec, dashboard, brief) · Feedback loop

2.2 ANSWERING **Rank the AI investments. What would you build first?**

Build one for quality, one for adoption.

The spec generator makes experiments better; the dashboard makes progress **visible**. Adoption and buy-in decide ROI, not cleverness — if you don't track it and show it, it's no use. The tempting one is deliberately last.

1	**AI experiment specification generator**	Biggest bite out of the 60%. Turns a hypothesis plus history into a powered, reviewable spec.	BUILD NOW
2	**AI dashboard summarisation**	The adoption engine. Visible, plain-English results win buy-in and drive change management. If it isn't shown, it isn't used.	BUILD NOW
3	**AI learning-brief generator**	The loop-closer: turns each result into graded knowledge in the evidence base. Low risk.	SEQUENCE NEXT
4	**AI QA assistant for launches**	The safety net that would have caught the German currency error. Fast-follow.	SEQUENCE NEXT
5	**AI promotion recommendation engine**	Most powerful, highest risk. It's what failed in Part 3. Needs the data, guardrails & QA first.	EARNS IT LATER
6	**AI customer-persona simulator**	Decision-support with confirmation-bias risk. Build after the loop is trusted.	EARNS IT LATER

2.3 ANSWERING **Design the knowledge-management system.**

Not a document store. A learning loop that grades its own evidence.

Simplest version that works: store every experiment as a graded record, keep one living belief per segment, and let two agents read and write it. The evidence base in the middle is the only thing that compounds.

`raw/`

THE LEDGER

Every experiment as a structured record: holdout, sample size, incrementality + confidence, cost, decision, caveats. Append-only; each record carries an evidence grade.

→ **Analytics agent** grade · update · flag

`context/`

THE EVIDENCE BASE · THE ASSET

One living belief per market × segment × offer: current effect, uncertainty, grade and last-verified date. This is what compounds.

A holdout › B flawed › C observed › D hunch

→ **Spec agent** retrieve · draft

`views/`

AUTO-GENERATED, ALWAYS IN SYNC

Dashboard, learning note and draft spec, rendered from the evidence base, never hand-written. Cheap, disposable, always current.

⚡ **The rule that would have caught Part 3:** a grade-D hunch can never overrule a grade-A holdout. Evidence quality decides, not recency or seniority.

Humans validate: a new belief lands as **candidate** → an analyst signs off and grades it → **confirmed**. Contradictions are flagged for review, never silently overwritten.

Measured by decision quality, not pages: did retrieved evidence change a call, was it right, is it still fresh. Plain files you own, not locked in a vendor.

Five personas, built from behaviour, not stereotypes.

What a data-built profile that predicts how a market reacts. **How** from booking + holdout behaviour, not clichés. **Why** to pressure-test an offer before spending a scarce slot, always an input to a real test, never the verdict.

UK — Deal-aware, high basket

Very incremental first (71%), wasteful repeat (31%).

Use for: Where to stop discounting loyal buyers first.

FR — Read with caution

Targeting issue makes 58% directional, not fact.

Use for: What to fix before we test here.

DE — Framing-sensitive

Strong first-purchase; prefers fixed value (qual).

Use for: Pre-test fixed vs % at equal cost.

IT — Read with caution

Flagged; low basket (€79) shifts fixed-offer maths.

Use for: Sequencing: measure first, spend later.

ES — Clean but marginal

48% on the lowest basket (€75).

Use for: How shallow a discount still acquires.

HOW I KEEP THEM HONEST

- Every trait traces to a booking or holdout number, not a stereotype.
- Log persona predictions vs live results and recalibrate. A persona that always agrees is a red flag.

WHEN I IGNORE IT

- When a live holdout contradicts it. Real data always wins.
- On flagged markets, novel offers, or high-stakes one-way doors. It prompts a test, it doesn't replace one.

2.5 ANSWERING **How would you measure success at six months?**

Different metrics matter at different stages.

Time saved is the hypothesis that wins the go-ahead; **quality** proves it works; **adoption** proves it landed; **ROI** proves it was worth running.

1 BEFORE
the hypothesis
Time saved

- Analyst hours per experiment reclaimed
- Idea-to-launch time (–60% target)
- The business case that wins the go-ahead

2 DURING BUILD
does it work?
System quality

- Spec accuracy & acceptance rate
- Experiment quality (reach significance)
- QA catch rate; recommendation hit-rate

3 AT LAUNCH
did it land?
Adoption

- Active users & experiments run through it
- Rests on UX + enablement + change mgmt
- Then: stakeholder satisfaction

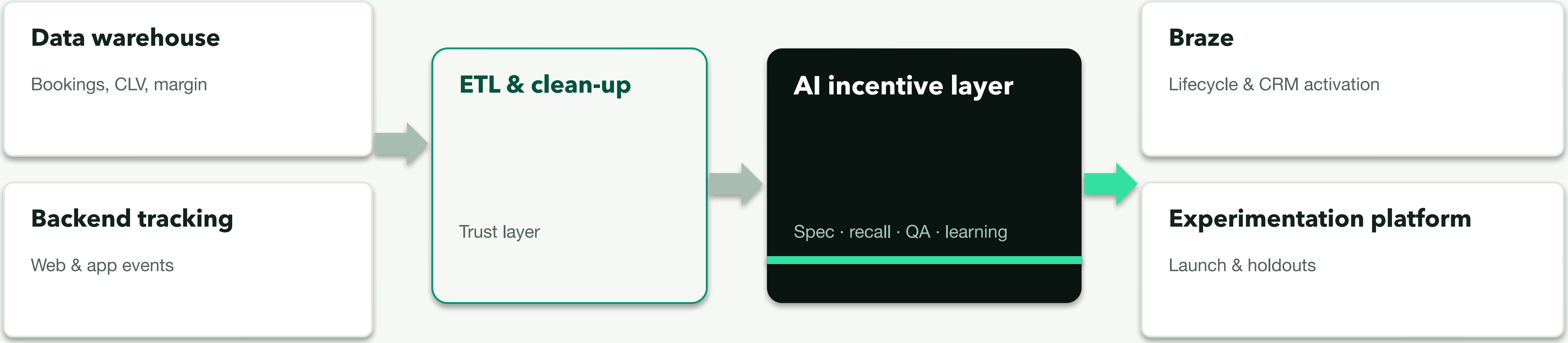
4 6 MONTHS
was it worth it?
Revenue & ROI

- Net incremental revenue per £ of promo
- Spend cut while bookings hold
- ROI net of run-cost (AI isn't free to run)

Adoption is the real leading indicator: a technically great system nobody uses returns nothing, which is why UX and enablement sit inside the metric, not beside it.

Integration, not islands.

The value only lands if this plugs into the systems teams already run. A standalone tool gets ignored. So the AI layer sits in the middle of the existing stack, reading clean data and writing back into the tools where work happens.



The unsexy truth: most of the build is connectors, schema and permissions, not prompts. I'd scope those dependencies on day one, because in a lean org an un-owned integration is where projects quietly die.

Why this won't be “a chatbot”.

A generic prompt box is too vague to be used properly, so it gets abandoned. Adoption comes from AI that does one job, for one role, inside the tool they already work in.

× A GENERIC CHATBOT

Vague, blank-box prompts
No workflow, no context
“Interesting” once, then ignored
Trust never builds

✓ EMBEDDED, PER ROLE, PER WORKFLOW

Analyst	gets a drafted, powered spec inside the experiment tool.
Marketer	gets a vetted recommendation inside Braze, where they build sends.
Leadership	gets the plain-English learning brief inside the dashboard.

P3 ANSWERING **Give 5+ reasons the AI recommendation failed.**

The engine swapped 15% for a €10 offer in Germany. It flopped. Why?

First suspect: a flat €10 is **less** than the €15.30 that 15% gave on a €102 basket, and it was even quoted in pounds. It under-priced the offer.

1 Currency & basket mismatch

Investigate → Recompute effective discount vs the 15% baseline; check the currency field (£ vs €).

2 Applied to the wrong segment

Investigate → Break the live result by segment; confirm who received it vs where the signal came from.

3 Soft signal as a hard rule

Investigate → Pull the qual study's sample size, date and segment; test where the preference holds.

4 Correlation, not incrementality

Investigate → Audit the objective & features; re-rank the pick on holdout incrementality only.

5 No clean baseline

Investigate → Confirm a holdout existed; check seasonality, competitor promos and tracking breaks.

6 Confident on thin data

Investigate → Check training-data coverage for fixed-value offers and whether a confidence score was ignored.

The lesson: guardrails before autonomy. The QA assistant and an incrementality-first knowledge base would have caught this. That's why they rank above the engine.

P4 ANSWERING **Align Finance and the Incentives Lead.**

Finance wants proof. The Lead wants the waste gone. Both are right.

FINANCE

“It's delivering bookings. I'm uncomfortable reducing discounts before we have stronger evidence.”

INCENTIVES LEAD · PRIVATELY

“We're wasting promotional budget on customers who would have booked anyway.”

MY MOVE · TEST ON A SAMPLE · DECIDE ON THE NUMBERS · REPORT TO BOTH

- **Sample, don't switch.** A holdout on repeat buyers in 1–2 markets caps the downside. The promo runs everywhere else, so Finance's bookings stay protected.
- **Pre-agree the decision rule.** We write down now what result cuts, keeps or scales the discount, so evidence decides, not opinion or seniority.
- **One live scorecard, both in the loop.** Finance and the Lead see the same weekly read of spend, bookings and incrementality. Visibility replaces the debate.

Land it with small wins, not a big-bang rollout.

A large, resource-constrained org with shifting priorities rewards proof over promises. So I'd earn trust and budget in stages, keeping dependencies small at each step.

WEEKS 0–4

Prove the data

Data clean-up + re-run the France/Italy holdout. First visible win: numbers people trust.

No engineering ask yet

WEEKS 4–10

Prove the loop

Ship the spec + learning-brief generator with one friendly team. One experiment, start to finish.

1 engineer, 1 team

WEEKS 10–20

Prove the value

Show time-to-launch down and a saving from the repeat-customer holdout. Take it to Finance.

Exec sponsor, scorecard

QUARTER 2+

Expand

Add QA, then earn the recommendation engine. Widen to more markets and teams.

Wider resource, on evidence

THE ONE IDEA

The flat 15% isn't a discount strategy. It's an un-tested assumption running across nine segments.

My whole submission is one move: make the assumption testable, build the loop that learns faster than we can argue, and land it with the data, integration, UX and buy-in that make it stick.

TO START, I'D NEED

✓ Access to warehouse & experiment data ✓ One engineer for the loop ✓ An exec sponsor + a friendly first team